

Robot Perception and Learning

Markov Decision Processes, Policy Iteration, Dynamic Programming, Monte Carlo Method, Temporal Difference

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Fall 2026



Appendix 1

Estimate Values of Actions by Incremental Implementation

- Let Q_n denote the estimate of its action value after being selected $n - 1$ times

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- Let's start rewriting Q_n :

$$\begin{aligned} Q_{n+1} &= \frac{1}{n} \sum_{i=1}^n R_i \\ &= \frac{1}{n} \left(R_n + \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) \frac{1}{n-1} \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) Q_n \right) \\ &= \frac{1}{n} \left(R_n + n Q_n - Q_n \right) \\ &= Q_n + \frac{1}{n} \left[R_n - Q_n \right], \end{aligned}$$

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Error

$\text{NewEstimate} \leftarrow \text{OldEstimate} + \text{StepSize} \times [\text{Target} - \text{OldEstimate}]$

Doesn't this look familiar? This is a standard form for learning/update rules!

Wait! We Assumed Stationary Reward Distributions

What if Reward Distributions are Nonstationary

- A distribution is **stationary** if it is fixed over time, otherwise, it is **nonstationary**
- If a distribution is nonstationary, we should trust recent rewards more than long-past rewards.
- We can re-write the incremental implementation with a step-size parameter $\alpha \in (0,1]$, which denotes how much we emphasize on recent rewards.

$$Q_{n+1} \doteq Q_n + \alpha [R_n - Q_n]$$

Exponential Recency-Weighted Average Method

$$\begin{aligned}Q_{n+1} &= Q_n + \alpha[R_n - Q_n] \\&= \alpha R_n + (1 - \alpha)Q_n \\&= \alpha R_n + (1 - \alpha)[\alpha R_{n-1} + (1 - \alpha)Q_{n-1}] \\&= \alpha R_n + (1 - \alpha)\alpha R_{n-1} + (1 - \alpha)^2 Q_{n-1} \\&= \alpha R_n + (1 - \alpha)\alpha R_{n-1} + (1 - \alpha)^2 \alpha R_{n-2} + \\&\quad \dots + (1 - \alpha)^{n-1} \alpha R_1 + (1 - \alpha)^n Q_1 \\&= (1 - \alpha)^n Q_1 + \sum_{i=1}^n \alpha (1 - \alpha)^{n-i} R_i.\end{aligned}$$

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How fast a term is forgotten

Appendix 2

Gradient Bandit Method

- Define $H_t(a_k)$ which characterizes relative preference of action a_k over other actions at time step t
- The probability for sampling an action is based on its preference value:

$$\Pr\{A_t = a\} \doteq \frac{e^{H_t(a)}}{\sum_{b=1}^k e^{H_t(b)}} \doteq \pi_t(a),$$

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- On each step, after selecting action A_t and receiving the reward R_t , the action preferences are updated by

$$\begin{aligned} H_{t+1}(A_t) &\doteq H_t(A_t) + \alpha(R_t - \bar{R}_t)(1 - \pi_t(A_t)), & \text{and} \\ H_{t+1}(a) &\doteq H_t(a) - \alpha(R_t - \bar{R}_t)\pi_t(a), & \text{for all } a \neq A_t, \end{aligned}$$

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- $\bar{R}_t$ is called **baseline**, with which an action is compared to
 - a_k is preferred if its reward R_t surpasses the baseline
- We can define $\bar{R}_t$ as the average of all rewards.

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- On each step, after selecting action A_t and receiving the reward R_t , the action preferences are updated by
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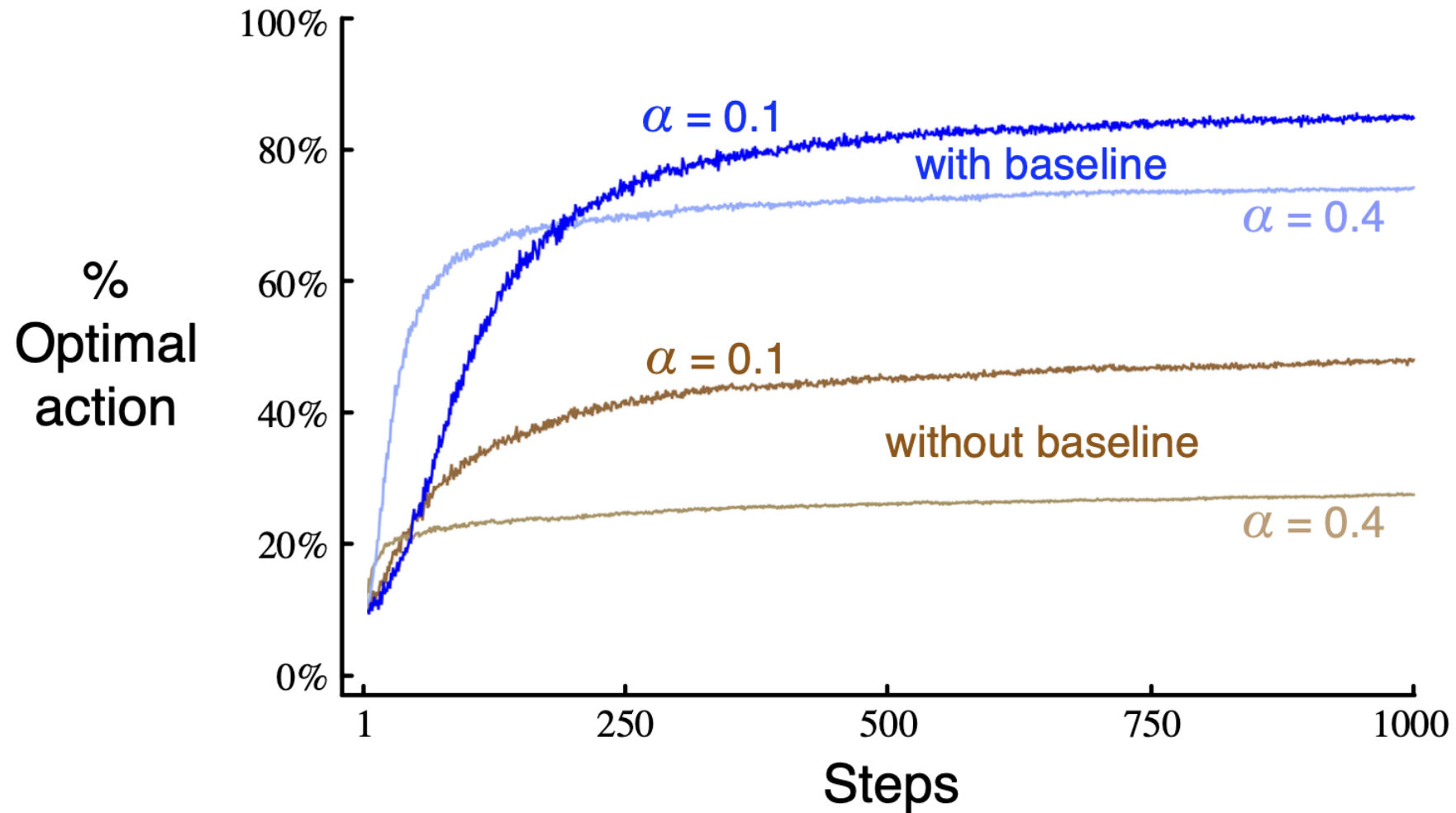
$$H_{t+1}(A_t) \doteq H_t(A_t) + \alpha(R_t - \bar{R}_t)(1 - \pi_t(A_t)), \quad \text{and}$$

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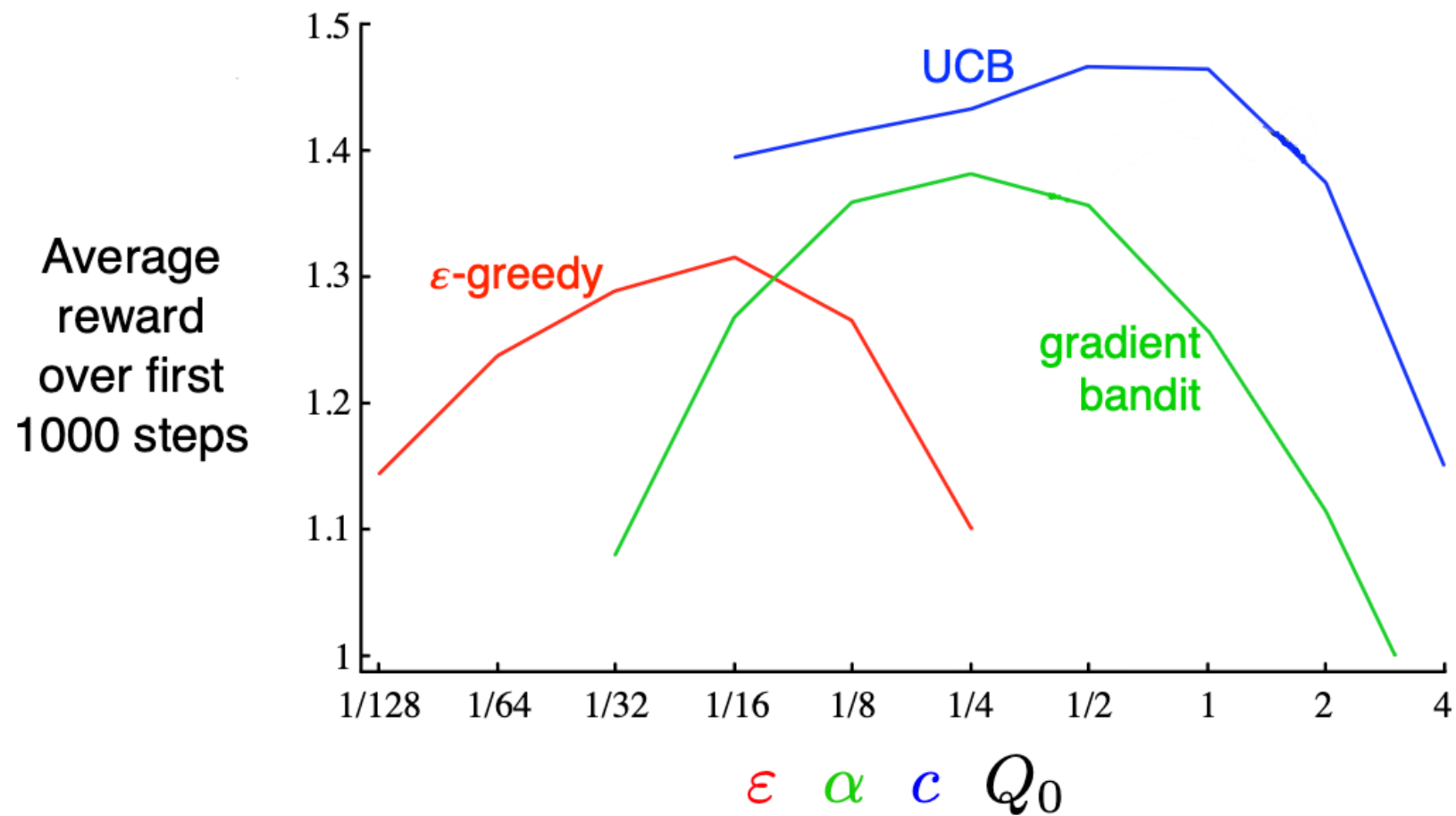
the preference of other actions should be largely decreased if they were frequently picked

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Gradient Bandit Method



Comparison of Three Methods



Appendix 3

Value Functions are Expected Returns

- The action value function $q_\pi(s, a)$: the expected return starting from state s and action a following policy π

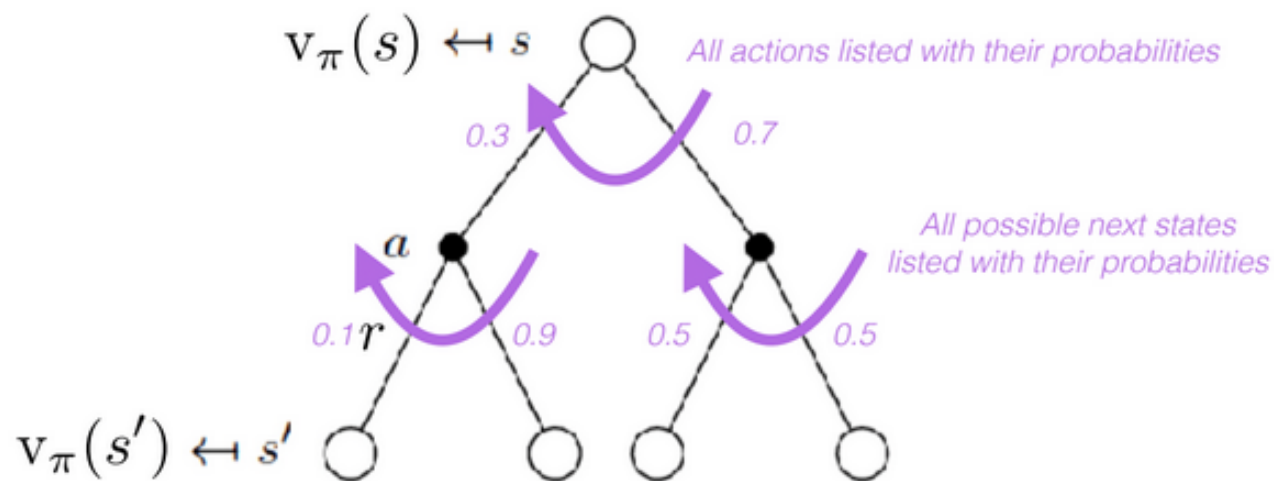
$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a] = \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')]$$

- The action-value function $q_\pi(s, a)$ estimates the “goodness” of action a at state s when deploying policy π . Simply speaking, how easy it is to achieve the goal with action a at state s .
- It's obvious to see

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')] = \sum_a \pi(a|s) q_\pi(s, a)$$

Back-up diagram for value functions

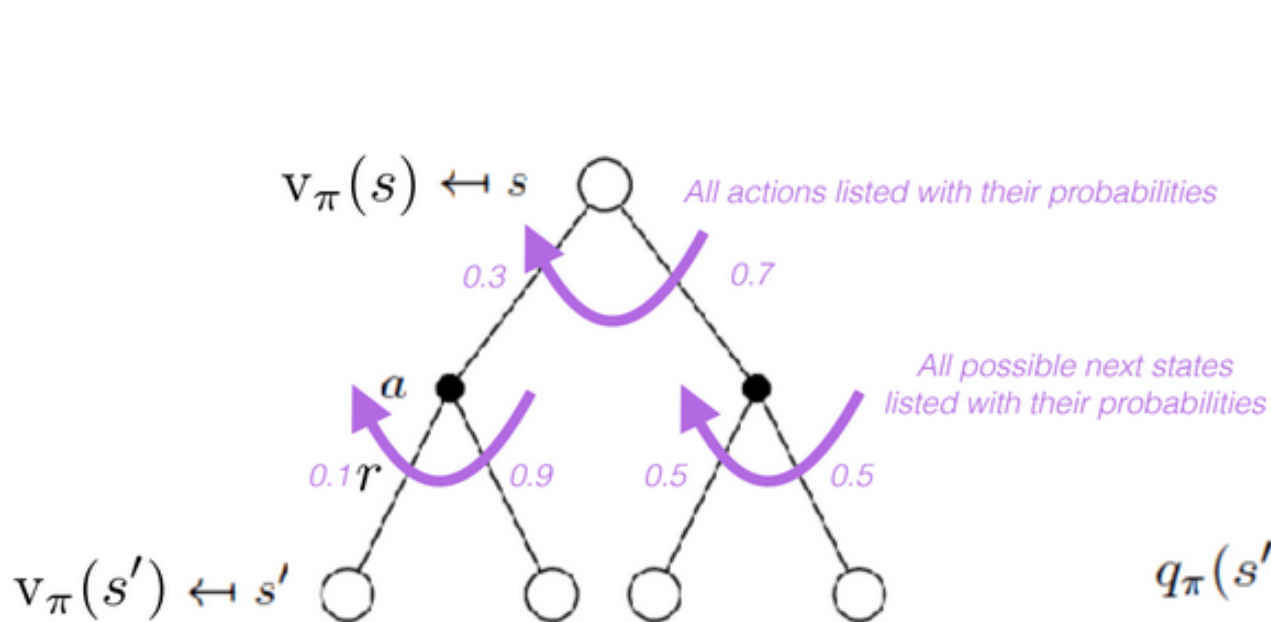
The probabilities of landing on each of the leaves sum to 1



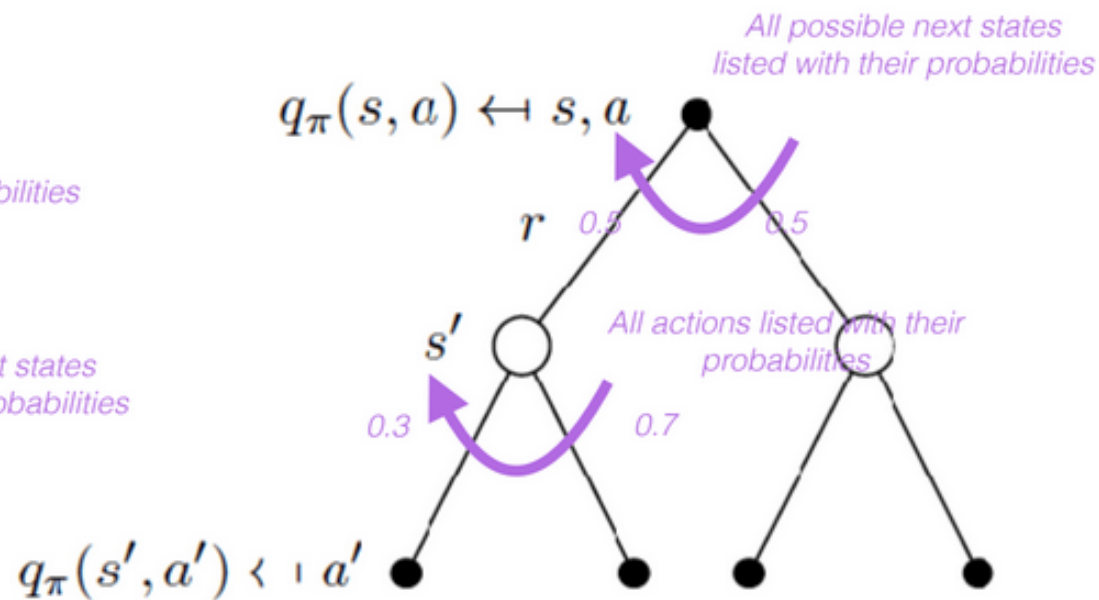
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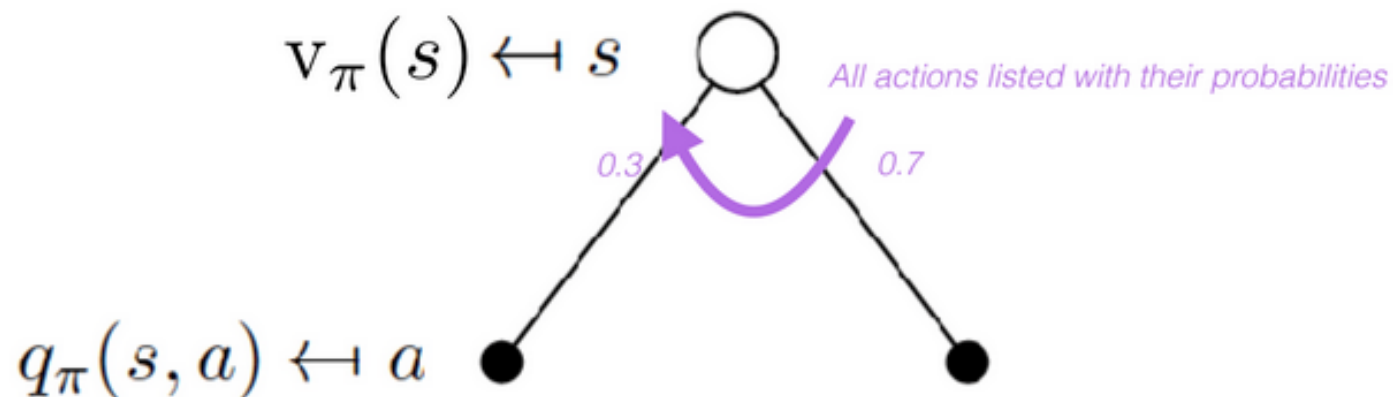


$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v_\pi(s')]$$



$$q_\pi(s, a) = \sum_{r,s'} p(s',r|s,a) \left(r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a') \right)$$

Relating state and state/action value functions



$$V_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a | s) q_{\pi}(s, a)$$

Appendix 4

We Have the Optimal Policy if We Know $q^*(s, a)$

- An optimal policy can be found by maximising over $q^*(s, a)$:

$$\pi^*(a|s) = \begin{cases} 1, & \text{if } a = \underset{a}{\operatorname{argmax}} q^*(s, a). \\ 0, & \text{otherwise.} \end{cases}$$

- An optimal policy can be found by maximizing over $v^*(s)$ with one-step look ahead:

$$\pi^*(a|s) = \begin{cases} 1, & \text{if } a = \underset{a}{\operatorname{argmax}} (\sum_{s', r} p(s', r|s, a))(r + \gamma v_*(s')) \\ 0, & \text{otherwise.} \end{cases}$$

Bellman Optimality Equation for v^*

1. We have $v^*(s) = \max_a q_{\pi^*}(s, a)$

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Why?

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_{\pi}(s, a)$$
$$\Rightarrow v^*(s) = \sum_{a \in \mathcal{A}} \pi^*(a|s) q_{\pi^*}(s, a) = \max_a q_{\pi^*}(s, a)$$

$$\pi^*(a|s) = \begin{cases} 1, & \text{if } a = \operatorname{argmax}_a q^*(s, a). \\ 0, & \text{otherwise.} \end{cases}$$

Bellman Optimality Equation for v^*

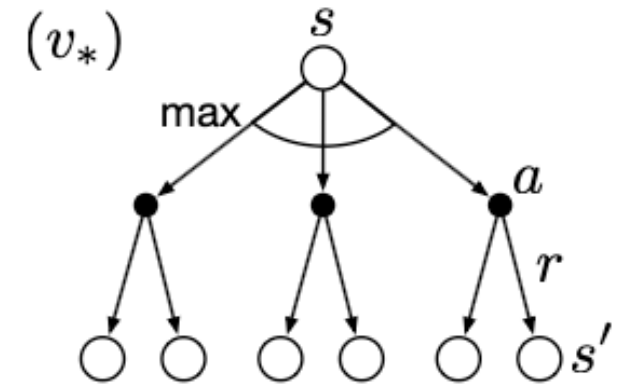
1. We have $v^*(s) = \max_a q_{\pi^*}(s, a)$
2. We can further write the equation:

$$\begin{aligned} v^*(s) &= \max_a q_{\pi^*}(s, a) \\ &= \max_a \mathbb{E}_{\pi^*}[G_t | S_t = s, A_t = a] \\ &= \max_a \mathbb{E}_{\pi^*}[R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a] \\ &= \max_a \mathbb{E}[R_{t+1} + \gamma v^*(S_{t+1}) | S_t = s, A_t = a] \\ &= \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v^*(s')] \end{aligned}$$

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This is again the recursive relationship!

Bellman Optimality Equation for q^*

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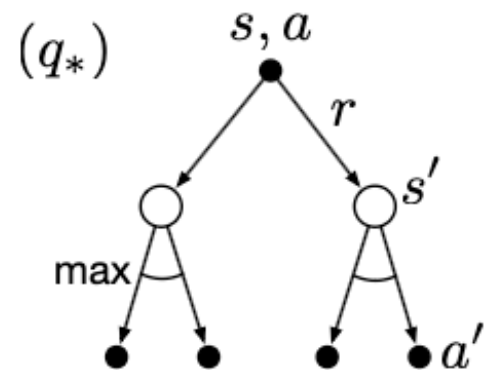
$$\begin{aligned} q^*(s, a) &= \mathbb{E}_{\pi^*}[R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a] \\ &= \mathbb{E}[R_{t+1} + \gamma v^*(S_{t+1}) | S_t = s, A_t = a] \\ &= \mathbb{E}\left[R_{t+1} + \gamma \max_{a'} q_{\pi^*}(S_{t+1}, a') \mid S_t = s, A_t = a\right] \\ &= \sum_{s', r} p(s', r | s, a) [r + \gamma \max_{a'} q^*(s', a')] \end{aligned}$$

Bellman Optimality Equation for q^*

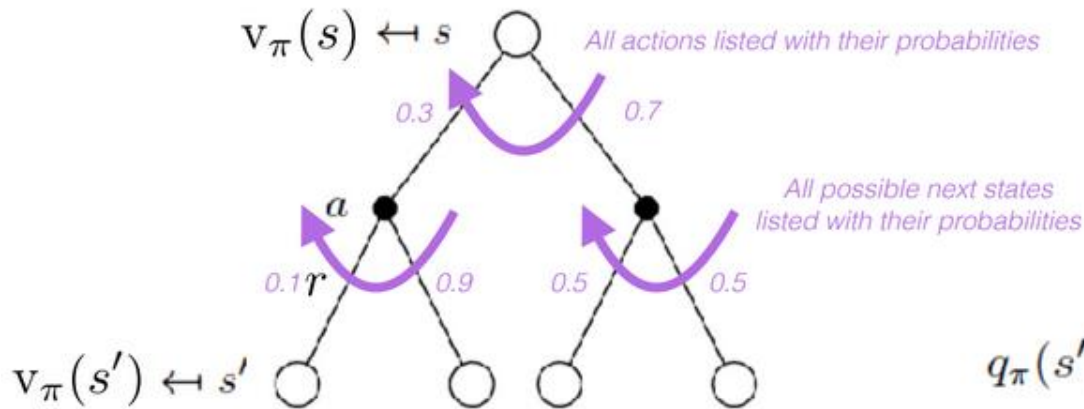
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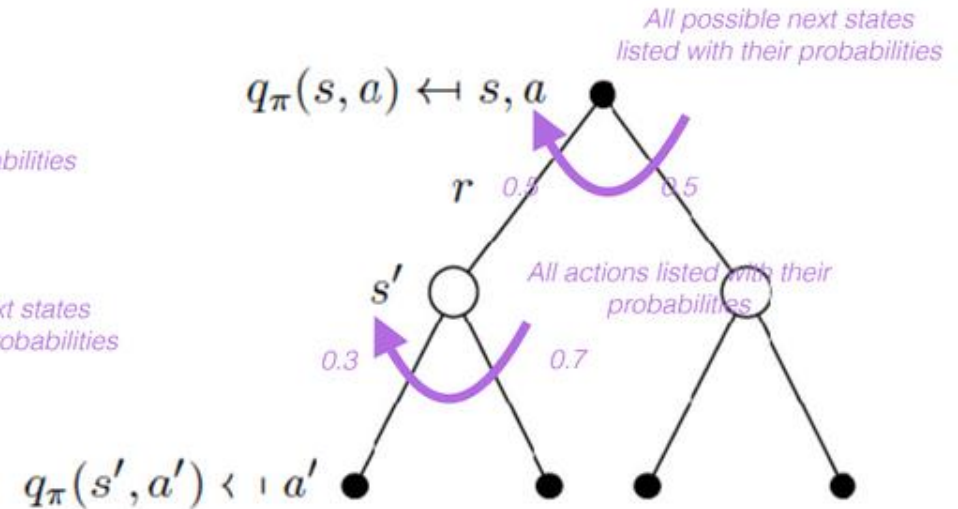
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Back-up Diagrams for Value Functions

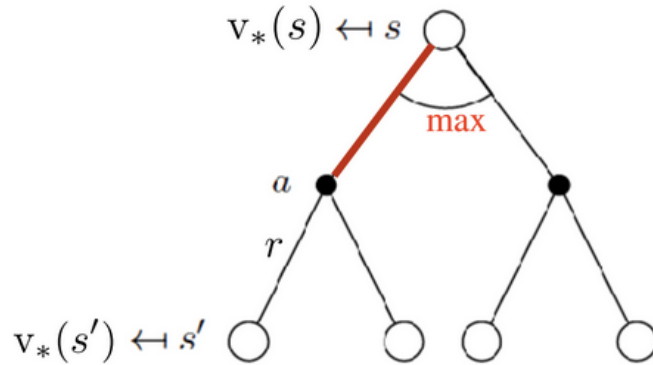


$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v_\pi(s')]$$



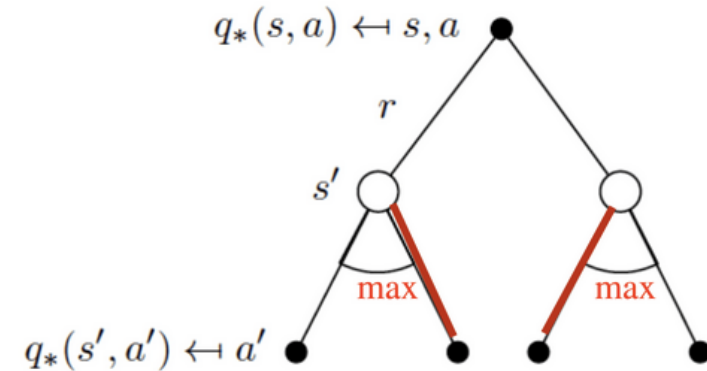
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Back-up Diagrams for Optimal Value Functions



For the Bellman expectation equations we sum over all the leaves, here **we choose only the best action branch!**

$$v_*(s) = \max_{a \in \mathcal{A}} \left(\sum_{s', r} p(s', r | s, a) (r + \gamma v_*(s')) \right)$$

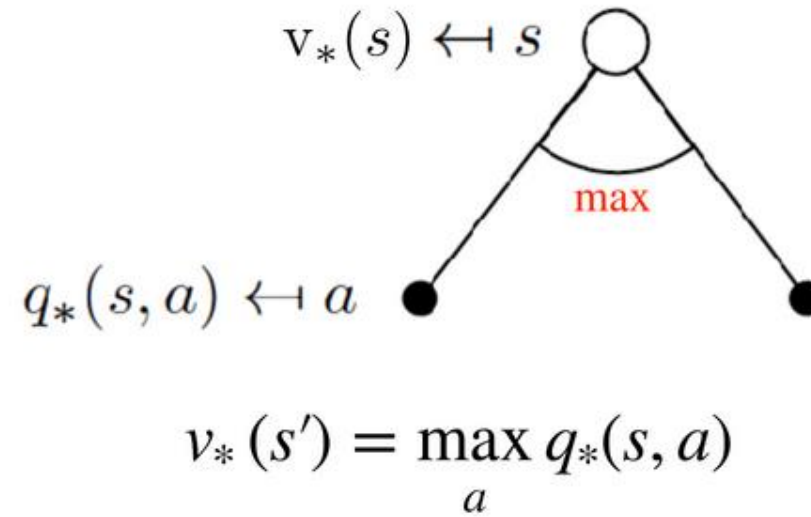


$$q_*(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a' \in \mathcal{A}} q_*(S_{t+1}, a') | S_t = s, A_t = a]$$

$$= \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} q_*(s', a') \right]$$

q^* is the unique solution of this system of nonlinear equations

Relating Optimal State and Action Value Functions



Appendix 5

In fact, the State Value Function following the Greedy Policy π' is Better than Original Policy π

$$\begin{aligned} v_{\pi}(s) &\leq q_{\pi}(s, \pi'(s)) \\ &= \mathbb{E}[R_{t+1} + \gamma v_{\pi}(S_{t+1}) \mid S_t = s, A_t = \pi'(s)] \\ &= \mathbb{E}_{\pi'}[R_{t+1} + \gamma v_{\pi}(S_{t+1}) \mid S_t = s] \end{aligned}$$

$$\pi'(a|s) = \begin{cases} 1, & \text{if } a = \underset{a}{\operatorname{argmax}} (\sum_{s',r} p(s',r|s,a))(r + \gamma v_{\pi}(s')) \\ 0, & \text{otherwise.} \end{cases}$$

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$$= \mathbb{E}_{\pi'}[R_{t+1} + \gamma v_{\pi}(S_{t+1}) \mid S_t = s]$$

$$\leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma q_{\pi}(S_{t+1}, \pi'(S_{t+1})) \mid S_t = s]$$

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Appendix 6

Proof of Optimality?

Definition 1. A Bellman optimality operator $\mathcal{T} : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$ is an operator that satisfies: for any $V \in \mathbb{R}^{|S|}$,

$$(\mathcal{T}V)(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim T(s'|s, a)} V(s')] .$$

Value iteration can thus be represented as recursively applying the Bellman optimality operator:

$$V_{k+1} = \mathcal{T}V_k. \tag{3}$$

The Bellman optimality operator $\mathcal{T}$ has several excellent properties. It is easy to verify that V^* is a fixed point of $\mathcal{T}$, i.e., $\mathcal{T}V^* = V^*$. Another important property is that $\mathcal{T}$ is a contraction mapping.

Proof of Optimality?

In finite dimensional coordinate space, let $x = (x_1, \dots, x_n)$:
 $\|x\|_\infty := \max(|x_1|, \dots, |x_n|)$

Theorem 2. $\mathcal{T}$ is a contraction mapping under sup-norm $\|\cdot\|_\infty$, i.e., there exists $\gamma \in [0, 1)$ such that

$$\|\mathcal{T}U - \mathcal{T}V\|_\infty \leq \gamma \|U - V\|_\infty, \forall U, V \in \mathbb{R}^{|S|}.$$

Theorem 4. Value iteration (3) converges to V^* , i.e.,

$$\lim_{k \rightarrow \infty} V_k = V^*,$$

where $V_k = \mathcal{T}^{k-1}V_0$.

Our goal is to show

Proof of Optimality?

Theorem 4. *Value iteration (3) converges to V^* , i.e.,*

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Proof. Note that V^* is a fixed point of $\mathcal{T}$. In addition, according to Theorem 2, $\mathcal{T}$ is a contraction mapping. Therefore,

$$\|V_k - V^*\|_\infty = \|\mathcal{T}V_{k-1} - \mathcal{T}V^*\|_\infty \leq \gamma \|V_{k-1} - V^*\|_\infty \leq \cdots \leq \gamma^k \|V_0 - V^*\|_\infty.$$

Let $k \rightarrow \infty$, and we have $\|V_k - V^*\|_\infty \rightarrow 0$. Thus $\lim_{k \rightarrow \infty} V_k = V^*$. □

We Need to Prove $\mathcal{T}$ is a Contraction Mapping

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$$\|\mathcal{T}U - \mathcal{T}V\|_\infty \leq \gamma \|U - V\|_\infty, \forall U, V \in \mathbb{R}^{|S|}.$$

Proof. To prove this property, we need the following lemma:

Lemma 3.

$$\left| \max_a f(a) - \max_a g(a) \right| \leq \max_a |f(a) - g(a)|.$$

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- Assume without loss of generality $\max_a f(a) \geq \max_a g(a)$ and denote $a^* = \operatorname{argmax}_a f(a)$

$$\left| \max_a f(a) - \max_a g(a) \right| = \max_a f(a) - \max_a g(a) = f(a^*) - \max_a g(a) \leq f(a^*) - g(a^*) \leq \max_a |f(a) - g(a)|.$$

Lemma 3.

$$\left| \max_a f(a) - \max_a g(a) \right| \leq \max_a |f(a) - g(a)|.$$

- Assume without loss of generality $\max_a f(a) \geq \max_a g(a)$ and denote $a^* = \operatorname{argmax}_a f(a)$

$$\left| \max_a f(a) - \max_a g(a) \right| = \max_a f(a) - \max_a g(a) = f(a^*) - \max_a g(a) \leq f(a^*) - g(a^*) \leq \max_a |f(a) - g(a)|.$$

- Let's prove theorem 2

$$\begin{aligned} |\mathcal{T}V(s) - \mathcal{T}U(s)| &= \left| \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim T(s'|s, a)} V(s')] - \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim T(s'|s, a)} U(s')] \right| \\ &\leq \max_a \left| \gamma \mathbb{E}_{s' \sim T(s'|s, a)} [V(s') - U(s')] \right| \\ &\triangleq \left| \gamma \mathbb{E}_{s' \sim T(s'|s, a^*)} [V(s') - U(s')] \right| \quad \text{where, } a^* \text{ is the argmax of the RHS above} \\ &\leq \gamma \max_{s'} |V(s') - U(s')| \\ &= \gamma \|V - U\|_\infty \end{aligned}$$

Lemma 3.

$$\left| \max_a f(a) - \max_a g(a) \right| \leq \max_a |f(a) - g(a)|.$$

- Let's prove theorem 2

$$\boxed{|\mathcal{T}V(s) - \mathcal{T}U(s)|} = \left| \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim T(s'|s, a)} V(s')] - \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim T(s'|s, a)} U(s')] \right|$$

Hold for any state s !

$$\begin{aligned} &\leq \max_a \left| \gamma \mathbb{E}_{s' \sim T(s'|s, a)} [V(s') - U(s')] \right| \\ &\triangleq \left| \gamma \mathbb{E}_{s' \sim T(s'|s, a^*)} [V(s') - U(s')] \right| \quad \text{where, } a^* \text{ is the argmax of the RHS above} \\ &\leq \gamma \max_{s'} |V(s') - U(s')| \\ &= \gamma \|V - U\|_\infty \end{aligned}$$

- Lemma 3 also holds when:

$$\max_s |\mathcal{T}V(s) - \mathcal{T}U(s)| \leq \gamma \|V - U\|_\infty, \quad \longrightarrow \quad \|\mathcal{T}U - \mathcal{T}V\|_\infty \leq \gamma \|V - U\|_\infty.$$

Appendix 7

Monte Carlo with Exploring Starts

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

$\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}, a \in \mathcal{A}(s)$

Loop forever (for each episode):

Choose $S_0 \in \mathcal{S}, A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0

Generate an episode from S_0, A_0 , following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$\pi(S_t) \leftarrow \arg \max_a Q(S_t, a)$

Monte Carlo with Exploring Starts

Converges to optimal policy, however, inefficient to start with every state-action pairs!

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:

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Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

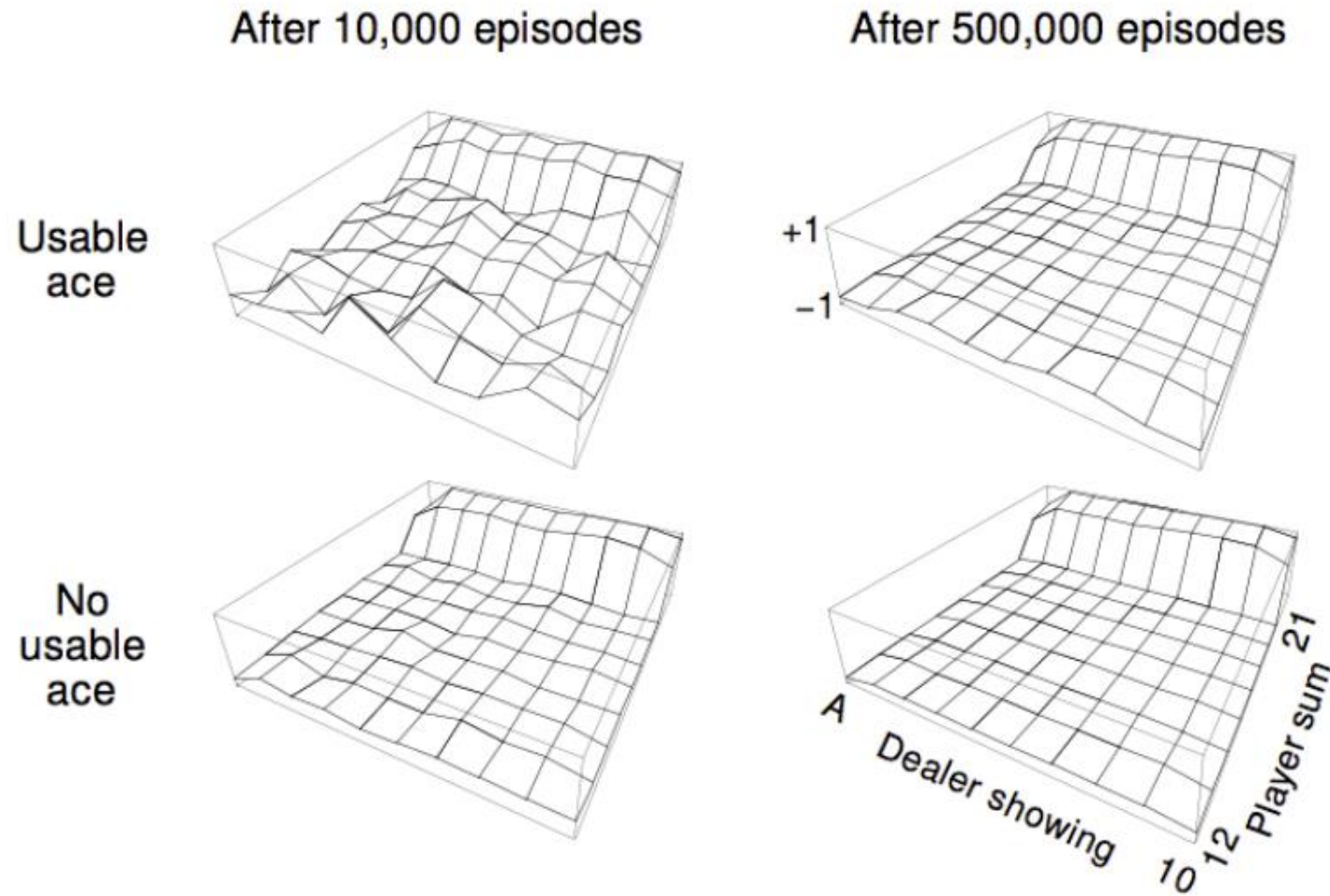
$\pi(S_t) \leftarrow \arg \max_a Q(S_t, a)$

The Blackjack Example

- States (200 of them):
 - Current sum (12-21)
 - Dealer's showing card (ace-10)
 - Do I have a "useable" ace? (yes-no)
- Action **stick**: Stop receiving cards (and terminate)
- Action **twist**: Take another card (no replacement)
- Reward for **stick**:
 - +1 if sum of cards $>$ sum of dealer cards
 - 0 if sum of cards = sum of dealer cards
 - -1 if sum of cards $<$ sum of dealer cards
- Reward for **twist**:
 - -1 if sum of cards $>$ 21 (and terminate)
 - 0 otherwise
- Transitions: automatically **twist** if sum of cards $<$ 12

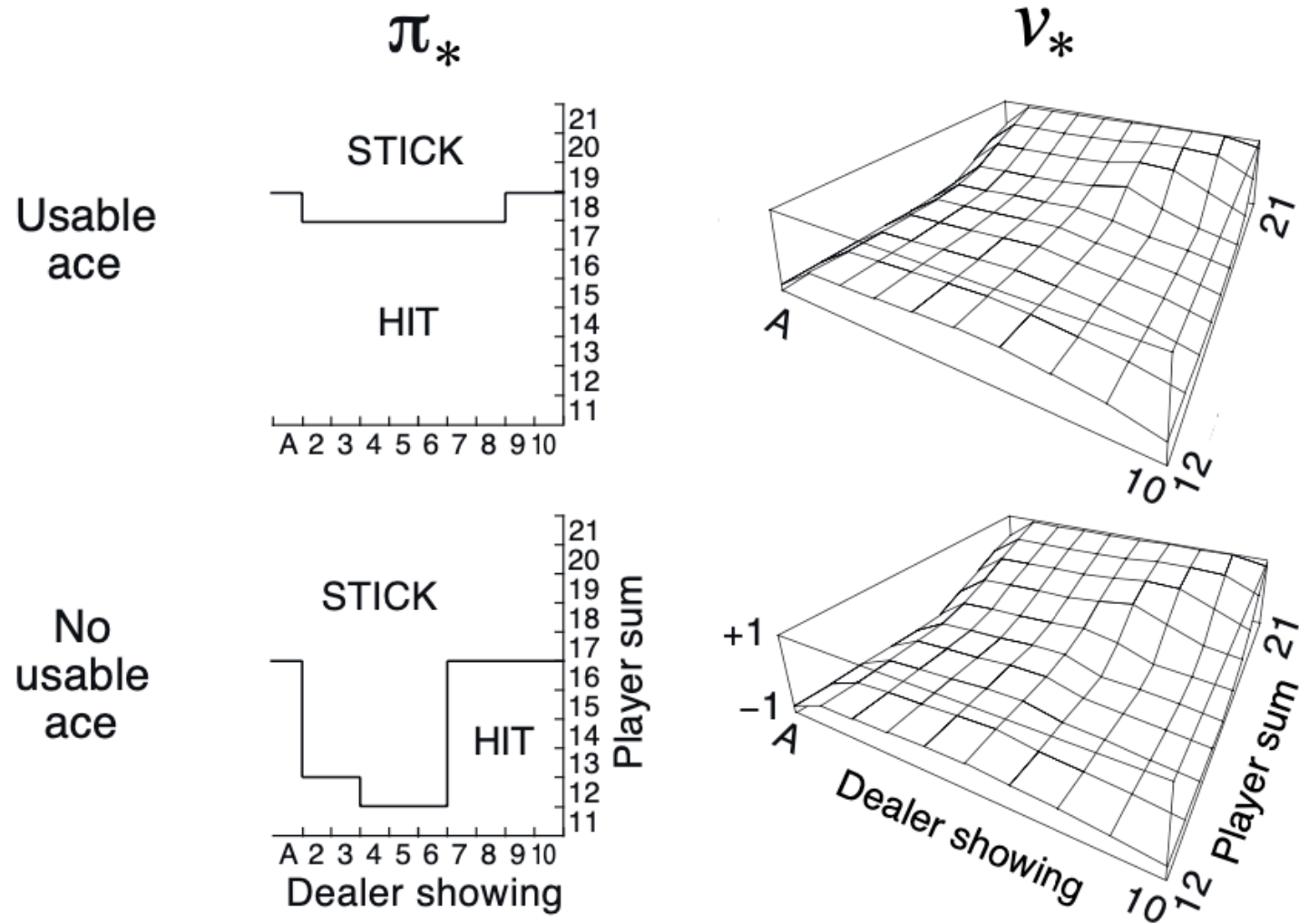


Value Estimate by a Monte Carlo Methods



Policy: **stick** if sum of cards ≥ 20 , otherwise **twist**

Optimal Policy found by Monte-Carlo ES



Monte Carlo Control with ϵ -soft Policies

Explore with probability $\frac{\epsilon}{A(s)}$ and exploit with probability $1 - \epsilon + \frac{\epsilon}{A(s)}$

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\epsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ϵ -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

ϵ -soft Policies Improve the Original Policy

$$\begin{aligned}\mathbb{E}_{\pi'} [q_{\pi}(s, \pi'(s))] &= \sum_a \pi'(a|s) q_{\pi}(s, a) \\&= \frac{\epsilon}{|\mathcal{A}(s)|} \sum_a q_{\pi}(s, a) + (1 - \epsilon) \max_a q_{\pi}(s, a) \\&\geq \frac{\epsilon}{|\mathcal{A}(s)|} \sum_a q_{\pi}(s, a) + (1 - \epsilon) \sum_a \frac{\pi(a|s) - \frac{\epsilon}{|\mathcal{A}(s)|}}{1 - \epsilon} q_{\pi}(s, a) \\&= \frac{\epsilon}{|\mathcal{A}(s)|} \sum_a q_{\pi}(s, a) - \frac{\epsilon}{|\mathcal{A}(s)|} \sum_a q_{\pi}(s, a) + \sum_a \pi(a|s) q_{\pi}(s, a) \\&= v_{\pi}(s).\end{aligned}$$

The Exploration-Exploitation Dilemma

- **Exploitation:** maximize the current highest reward: $\pi(s) = \underset{a}{\operatorname{argmax}} q_{\pi}(s, a)$
- **Exploration:** maximize the information about the environment
- **ALL learning methods faces the dilemma:** learning state-action values conditions on subsequent optimal behaviors but they need to act **sub-optimally** to explore all state-action pairs
- **Solutions:** Both methods are compromises. They learn action values not for the optimal policy, but for a near-optimal policy that still explores.
 - exploring starts: Every state-action pair has a non-zero probability of being the starting pair
 - ϵ -soft policies: Most of the time choose the action with maximal estimated action values, but with probability ϵ select a random action
 - off-policy: use different policies for collecting experience and evaluating

The Exploration-Exploitation Dilemma

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 - off-policy: use different policies for collecting experience and evaluating
- **ALL learning methods faces the dilemma:** learning state-action values conditions on subsequent optimal behaviors but they need to act **sub-optimally** to explore all state-action pairs
 - Let's have two policies: policy **b** to explore, and policy **π** to behave optimally

On-Policy and Off-Policy Learning

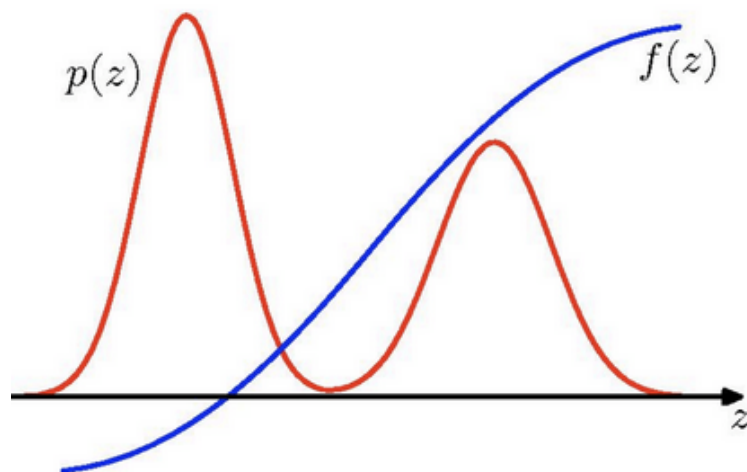
- On-policy learning: learn v_π and q_π for policy π that executes and explores
- Off-policy learning: learn v_π and q_π for target policy π from experience collected by behavior policy b
- We only need coverage: every action taken under π is also taken, at least occasionally, under b

$$\pi(a|s) > 0 \text{ implies } b(a|s) > 0$$

- The goodness of decoupling target and behavior policy:
 - Learn from observing humans or other agents
 - Re-use experience generated from old policies
 - Learn about optimal policy while following exploratory policy
 - Learn about multiple policies while following one policy

Estimating Expectations

- General Idea: Draw independent samples $\{z^1, \dots, z^N\}$ from distribution $p(z)$ to approximate expectation:



$$\mathbb{E}[f] = \int f(z)p(z)dz \approx$$

$$\frac{1}{N} \sum_{n=1}^N f(z^n) = \hat{f}.$$

Note that: $\mathbb{E}[f] = \mathbb{E}[\hat{f}]$.

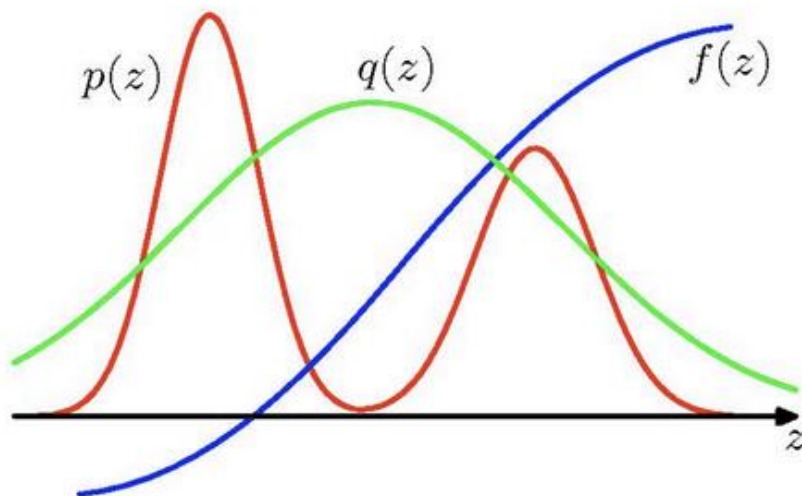
so the estimator has correct mean (unbiased).

- The **variance**: $\text{var}[\hat{f}] = \frac{1}{N} \mathbb{E}[(f - \mathbb{E}[f])^2]$.
- Variance decreases as $1/N$.
- **Remark:** The accuracy of the estimator **does not depend on dimensionality of z** .

Importance Sampling

- Suppose we have an **easy-to-sample proposal distribution $q(z)$** , such that

$$q(z) > 0 \text{ if } p(z) > 0. \quad \mathbb{E}[f] = \int f(z)p(z)dz$$

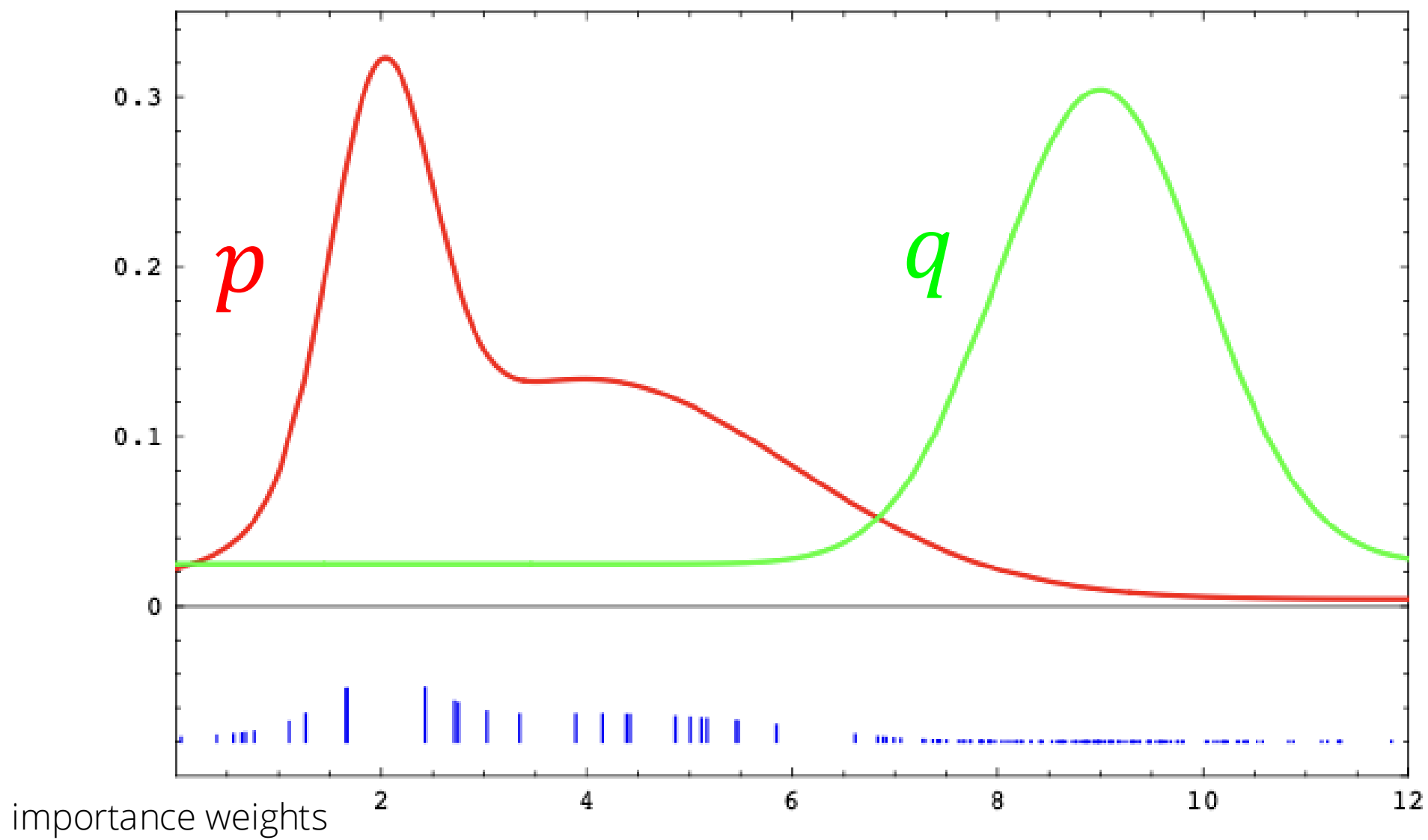


$$= \int f(z) \frac{p(z)}{q(z)} q(z) dz$$

$$\approx \frac{1}{N} \sum_n \frac{p(z^n)}{q(z^n)} f(z^n), \quad z^n \sim q(z).$$

- The quantities $w^n = p(z^n)/q(z^n)$ are known as importance weights.

- This is useful when we can evaluate the probability p but is hard to sample from it



Importance Sampling Ratio

- The probability of state-action trajectory $A_t, S_t, \dots, A_T, S_T$ under policy π :

$$\begin{aligned} & \Pr\{A_t, S_{t+1}, A_{t+1}, \dots, S_T \mid S_t, A_{t:T-1} \sim \pi\} \\ &= \pi(A_t|S_t)p(S_{t+1}|S_t, A_t)\pi(A_{t+1}|S_{t+1}) \cdots p(S_T|S_{T-1}, A_{T-1}) \\ &= \prod_{k=t}^{T-1} \pi(A_k|S_k)p(S_{k+1}|S_k, A_k), \end{aligned}$$

- The importance sampling ratio between target policy π and behavior policy b :

$$\rho_{t:T-1} \doteq \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k)p(S_{k+1}|S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k|S_k)p(S_{k+1}|S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}$$

- Estimate $v_\pi(s)$ from sampling with behavior policy b

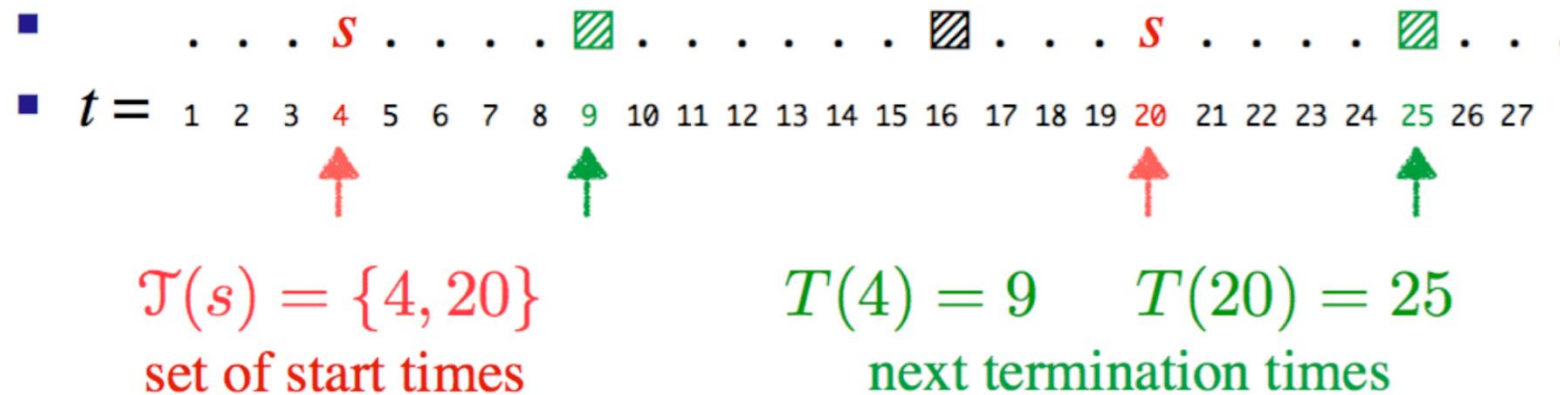
$$\mathbb{E}[\rho_{t:T-1}G_t \mid S_t=s] = v_\pi(s)$$

Importance Sampling

- Ordinary importance sampling forms estimate

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}.$$

- New notation: time steps increase across episode boundaries:



Importance Sampling

- Two ways of averaging weighted returns:
- Ordinary importance sampling forms estimate:

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}.$$

- Weighted importance sampling forms estimate:

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

Ordinary vs. Weighted Importance Sampling

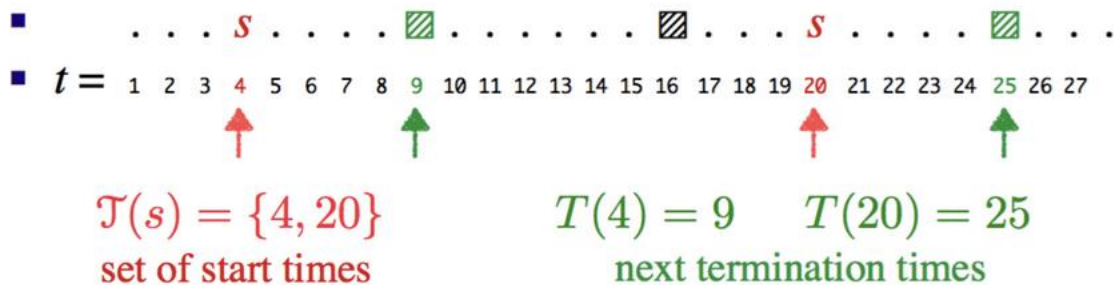
Ordinary Sampling:

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}$$

First time of termination following time t

return after t up through $T(t)$

Every time: the set of all time steps in which state s is visited



Weighted Sampling:

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

First-visit MC:

- Ordinary Sampling is unbiased, but the variance is unbounded
- Weighted Sampling is biased, but with much lower variance

Every-visit MC:

- Ordinary Sampling is biased
- Weighted Sampling is biased

Proof:

<https://link.springer.com/article/10.1007/BF00114726>

Importance Sampling

- Two ways of averaging weighted returns:

- Ordinary importance sampling forms estimate:

Correct mean of value following policy π

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}.$$

- Weighted importance sampling forms estimate:

These ratio in the numerator is cancelled by the denominator. The estimation is not correct anymore....

$$V(s) \doteq \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}$$

Remember the Incremental Implementation in the Bandit Problem

- Let Q_n denote the estimate of its action value after being selected $n - 1$ times

$$Q_n \doteq \frac{R_1 + R_2 + \cdots + R_{n-1}}{n - 1}$$

- Let's start rewriting Q_n :

$$\begin{aligned} Q_{n+1} &= \frac{1}{n} \sum_{i=1}^n R_i \\ &= \frac{1}{n} \left(R_n + \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) \frac{1}{n-1} \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) Q_n \right) \\ &= \frac{1}{n} \left(R_n + n Q_n - Q_n \right) \\ &= Q_n + \frac{1}{n} [R_n - Q_n], \end{aligned}$$

Incremental Implementation of Ordinary Importance Sampling

- We have:

$$V_n(s) = \frac{\rho_1 G_1 + \cdots \rho_{n-1} G_{n-1}}{n-1}$$

- We can re-write $V(s)$ as:

$$V_{n+1}(s) = V_n(s) + \frac{1}{n} [\rho_n G_n - V_n(s)]$$

Incremental Implementation of Weighted Importance Sampling

- We can re-write weighted importance sampling as:

$$V_n \doteq \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k}, \quad n \geq 2, \quad W_i = \rho_{t_i:T}(t_i) - 1$$

- The update rule is then:

$$V_{n+1} \doteq V_n + \frac{W_n}{C_n} [G_n - V_n], \quad n \geq 1,$$

$$C_{n+1} \doteq C_n + W_{n+1},$$

Off-Policy Monte Carlo Prediction

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

Loop forever (for each episode):

$b \leftarrow$ any policy with coverage of π

Generate an episode following b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$, while $W \neq 0$:

$G \leftarrow \gamma G + R_{t+1}$

$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Off-policy Monte Carlo Control

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$Q(s, a) \in \mathbb{R}$ (arbitrarily)

$C(s, a) \leftarrow 0$

$\pi(s) \leftarrow \operatorname{argmax}_a Q(s, a)$ (with ties broken consistently)

Loop forever (for each episode):

$b \leftarrow$ any soft policy

Generate an episode using b : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

$W \leftarrow 1$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

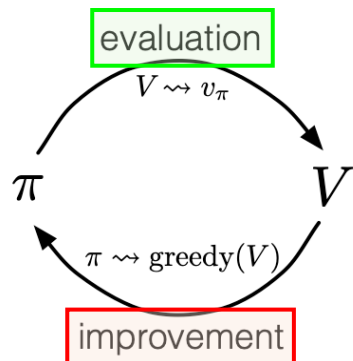
$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$

$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$

$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)

$W \leftarrow W \frac{1}{b(A_t|S_t)}$



Only learns from the tails of episodes (when all actions are greedy). Learning is very slow!

Check Section 5.8 and 5.9 of “Reinforcement Learning: An Introduction” for Discounting-aware Importance Sampling